

Coherent Algebraic Sheaves

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Chapter 1

Sheaves

1.1 Operations on Sheaves

1.1.1 Definition of a sheaf

Let X be a topological space. A *sheaf of abelian groups* on X (or simply a *sheaf*) consist of:

- (a) A function $x \rightarrow \mathcal{F}_x$ sending every $x \in X$ to an abelian group $\mathcal{F}_x$,
- (b) A topology on the set $\mathcal{F}$, the sum of the sets $\mathcal{F}_x$.

If f is an element of $\mathcal{F}_x$, we put $\pi(f) = x$; the map π is called the projection of $\mathcal{F}$ on X ; the subset of $\mathcal{F} \times \mathcal{F}$ formed the pairs (f, g) such that $\pi(f) = \pi(g)$ is denoted $\mathcal{F} + \mathcal{F}$.

With these definitions in place, we can state the two axioms to which the data (a) and (b) are subject:

- (I) For all $f \in \mathcal{F}$, there exist open neighborhoods V of f and U of $\pi(f)$ such that the restriction of π at V is a homeomorphism of V onto U